NORTH-WEST UNIVERSITY

CMPG 325 - COMPUTER NETWORKS

TECHNICAL REPORT

Project Title: Network Design and Implementation for

Kagisano-Molopo Local Municipality Offices (Ganyesa)

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Executive Summary

This technical report documents the complete design, implementation, and testing of a computer network for the Kagisano-Molopo Local Municipality Offices in Ganyesa, South Africa. The project was executed in two milestones: a design review phase and an implementation review phase.

The network follows a Core-Distribution-Access hierarchical topology serving seven distinct VLANs across five municipal departments. The assigned addressing block of 192.168.46.0/24 was efficiently subdivided using Variable Length Subnet Masking (VLSM) to support the Voice, Finance, Corporate Services, Community Services, Municipal Manager, Server Farm, and Management segments.

The assigned networking challenge DHCP with scoped multi-VLAN address assignment was successfully implemented using a centralised DHCP server and **ip helper-address** DHCP relay on the core router. Five DHCP scopes were configured, one for each user VLAN. All client devices received IP addresses from the correct pool.

The client's design constraint for a new application/file server within six months was addressed by provisioning a dedicated Server VLAN (VLAN 60) with a /28 subnet, ensuring seamless integration of future infrastructure without re-addressing existing VLANs.

The CR11 change request for VoIP handsets was fully accommodated through a dedicated Voice VLAN (VLAN 10), Cisco 7960 IP phones with 802.1Q tagging, and CallManager Express (CME) configured on the core router. All four phones registered successfully with extensions 1001 through 1004.

All 26 verification tests passed. The final network is stable, functional, and meets every client requirement.

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1. Introduction

1.1 Client Background

The Kagisano-Molopo Local Municipality Offices, located in Ganyesa in the North West Province of South Africa, serve as the administrative centre for local government service delivery. The municipality manages critical public services including budget and treasury operations, human resources, community development, infrastructure maintenance, and executive administration.

For a municipal office of this nature, network infrastructure must satisfy rigorous standards for service availability, security, and departmental isolation. Different departments handle different types of sensitive information financial records, personnel data, and citizen information that require logical separation.

1.2 Project Scope

This project required the design and simulation of a complete computer network for the municipality. The scope included:

- Analysis of client requirements across five departments
- Design of an appropriate physical and logical topology
- Development of an IP addressing plan based on the assigned block of 192.168.46.0/24
- Implementation of the network in Cisco Packet Tracer
- Configuration of the assigned networking challenge (DHCP scoped multi-VLAN address assignment)
- Accommodation of the client's design constraint (new server within six months)
- Accommodation of the change request CR11 (VoIP handsets)
- Comprehensive testing and verification
- Documentation of the entire process

1.3 Report Structure

This report follows a standard technical report structure. Section 2 analyses the client's requirements. Section 3 presents the network design. Section 4 details the implementation. Section 5 documents the testing and verification. Section 6 covers troubleshooting. Section 7 provides reflection and lessons learned. Section 8 concludes the report. Sections 9 and 10 provide references and appendices.

2. Client Requirements Analysis

2.1 Organisational Context

The Kagisano-Molopo Local Municipality Offices operate in the municipal services industry, providing essential public services to the Ganyesa community. The network must support day-to-day administrative operations while maintaining security boundaries between departments handling sensitive information.

2.2 Departmental Needs

The network was designed to serve five distinct departmental segments, each with specific requirements:

Table 1: Departmental Network Requirements

Department	VLAN	Primary Functions	Key Requirements
Voice	10	VoIP handsets	Low latency, prioritised traffic
Budget & Treasury	20	Municipal billing, revenue, payroll	High security, integrity
Corporate Services	30	HR records, legal documents	Confidentiality, throughput
Community & Technical	40	Infrastructure, social services	Stable Connectivity
Municipal Manager	50	Strategic planning, executive	Minimal latency, priority
Server Farm	60	Application/file servers	Centralised services
Management	99	Network device administration	Out-of-band access

Each department requires logical isolation from the others to prevent broadcast storms, enforce security boundaries, and support structured traffic control.

2.3 Assigned Addressing Block

The client was assigned the classless address block **192.168.46.0/24**, providing 256 total IP addresses. This block required efficient subdivision to accommodate all seven VLANs while reserving capacity for future growth.

2.4 Design Constraint

Stated Constraint: A new application or file server is planned within six months.

This constraint required the design to provision a dedicated, secure segment for the future server. Integrating servers into existing workstation subnets would compromise security and create addressing conflicts. Instead, a dedicated Server VLAN (VLAN 60) with its own subnet ensures the new server can be deployed without disrupting any existing configuration.

2.5 Change Request CR11

Change Request: Management adopts VoIP handsets - voice traffic must be accommodated on the existing design.

VoIP traffic is highly sensitive to packet loss, jitter, and delay. Placing IP handsets on a dedicated logical network allows implementation of Quality of Service (QoS) policies to prioritise voice traffic. This also allows physical efficiency: Cisco 7960 IP phones include a built-in three-port switch, allowing a workstation and IP phone to share a single physical switch port.

3. Network Design

3.1 Physical Topology

The Ganyesa Municipal facility is structured around a high-availability Hierarchical Star Topology based on the Core-Distribution-Access model.

Core Layer: A Cisco 2811 Integrated Services Router (CoreRouter) positioned at the root of the hierarchy. It performs inter-VLAN routing via router-on-a-stick, hosts DHCP relay agents for all user VLANs, and provides CallManager Express services.

Distribution Layer: A Cisco Catalyst 2960-24TT switch (DistSW) aggregates access-layer traffic and provides high-speed uplinks to the Core Router.

Access Layer: Five Cisco Catalyst 2960-24TT switches deployed one per departmental block, each running 802.1Q VLAN tagging and managing voice VLAN traffic on host ports.

Server Farm: A dedicated access switch (Server-ASW) hosts the centralised DHCP server on VLAN 60.

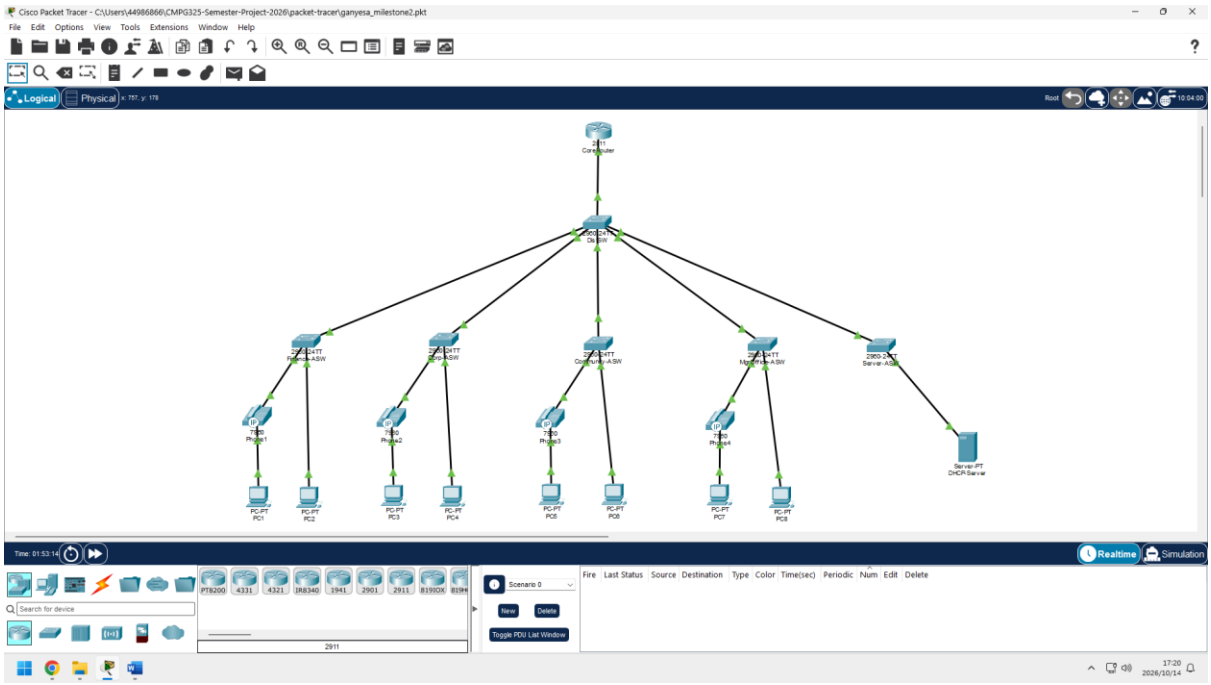


Figure 1: Physical topology - Core-Distribution-Access hierarchy

3.2 Logical Topology

The logical design segments the network into seven VLANs. Each VLAN forms its own broadcast domain, isolating traffic and enabling departmental security boundaries.

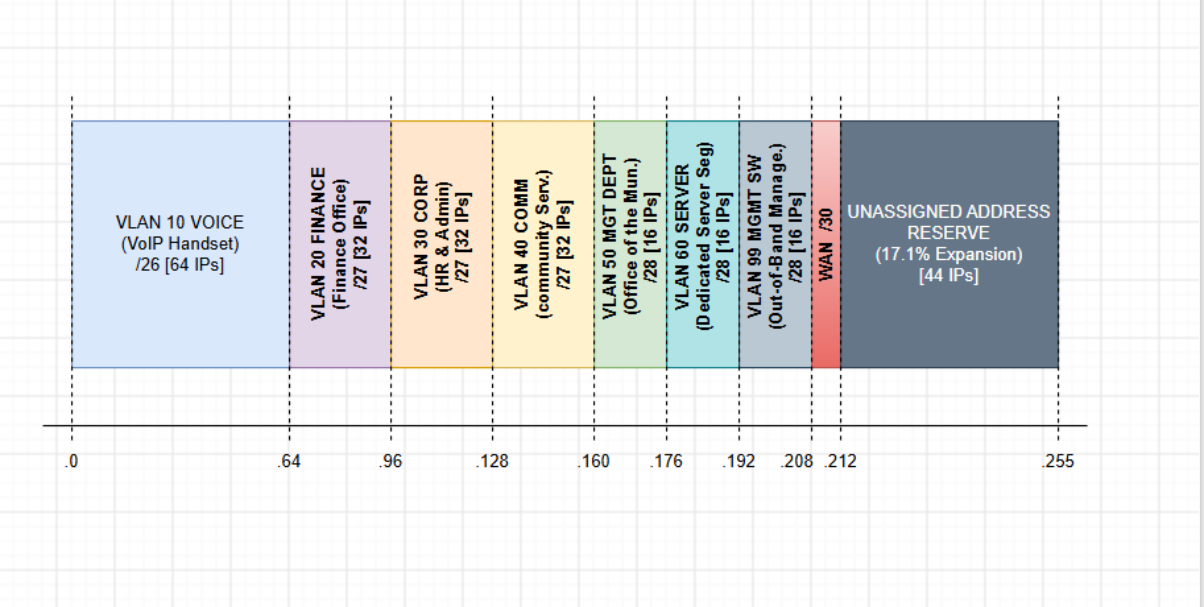


Figure 2: Logical VLAN and VLSM IP allocation map

VLAN assignment:

VLAN ID	Name	Purpose
10	VOICE	VoIP handsets
20	FINANCE	Budget & Treasury
30	CORP	Corporate Services
40	COMMUNITY	Community & Technical Services
50	MGT_DEPT	Municipal Manager
60	SERVER	Server Farm
99	MGMT_SWITCH	Out-of-band device management

3.3 IP Addressing Plan (VLSM)

To utilise the assigned /24 block efficiently, Variable Length Subnet Masking (VLSM) was applied. Subnets were allocated sequentially from the largest requirement down to the smallest, preserving a contiguous reserve for future growth.

Table 2: VLSM IP Addressing Plan

VLAN	Name	Network Address	Subnet Mask	Usable Range	Gateway	Broadcast Network
10	VOICE	192.168.46 .0 /26	255.255.255 .192	.1 - .62	192.168.46 .1	192.168.46 .63
20	FINANCE	192.168.46 .64 /27	255.255.255 .224	.65 - .94	192.168.46 .65	192.168.46 .95
30	CORP	192.168.46 .96 /27	255.255.255 .224	.97 - .126	192.168.46 .97	192.168.46 .127

40	COMMUNITY	192.168.46.128 /27	255.255.255.224	.129 - .158	192.168.46.129	192.168.46.159
50	MGT_DEPT	192.168.46.160 /28	255.255.255.240	.161 - .174	192.168.46.161	192.168.46.175
60	SERVER	192.168.46.176 /28	255.255.255.240	.177 - .190	192.168.46.177	192.168.46.191
99	MGMT_SWITCH	192.168.46.192 /28	255.255.255.240	.193 - .206	192.168.46.193	192.168.46.207
-	Reserve	192.168.46.208 /28+	-	.208 - .255	-	-

Reserve: 48 addresses (18.75% of the block) remain unassigned to support future expansion a new /27 subnet, multiple /28 segments, or WAN links.

3.4 Device Selection

Table 3: Device Inventory

Device	Model	Quantity	Role
CoreRouter	Cisco 2811	1	Router-on-a-stick, DHCP relay, CME
DistSW	Cisco 2960-24TT	1	Trunk aggregation
Access Switches	Cisco 2960-24TT	5	Departmental access
DHCP-Server	Server-PT	1	Centralised DHCP
PCs	PC-PT	8	Workstations
IP Phones	Cisco 7960	4	VoIP handsets

The Cisco 2811 was selected over the 2911 because Packet Tracer only supports CallManager Express (CME) on the 2811 model essential for the CR11 VoIP change request.

3.5 Design Justification

Hierarchical structure: The Core-Distribution-Access model isolates failures, simplifies routing, and optimises bandwidth. A failure at the access layer does not affect other departments.

Dedicated Voice VLAN: Voice traffic is separated for QoS and to ensure call quality is not impacted by data traffic bursts.

Dedicated Server VLAN: Aligns with the six-month server constraint. The new server can be added as a static host without re-addressing existing subnets.

VLSM over fixed-length subnetting: VLSM prevents IP wastage by right-sizing each subnet to its actual requirement. The Voice VLAN, for example, needs far fewer hosts than the general staff VLAN.

Centralised DHCP: A single DHCP server simplifies administration and reduces configuration errors. Combined with **ip helper-address**, it serves all VLANs across the routed network.

4. Implementation

4.1 Layer 2 Configuration

All six switches were configured with the seven required VLANs. Trunk ports were established between the distribution switch and all access switches using 802.1Q encapsulation with native VLAN 99.

Trunk configuration example (DistSW):

```
interface range FastEthernet0/1-5
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,50,60,99
```

Access ports with voice VLAN were configured on all departmental switches so that IP phones tag their own frames into VLAN 10, while the attached PC traffic remains in the department VLAN:

```
interface range FastEthernet0/1-2
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 10
 spanning-tree portfast
```

Each switch received a management IP on VLAN 99 for out-of-band administration (192.168.46.194 through .199).

```
DistSW>enable
DistSW#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/2
10   VOICE                            active
20   FINANCE                          active
30   CORP                             active
40   COMMUNITY                        active
50   MGT_DEPT                         active
60   SERVER                           active
99   MGMT_SWITCH                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
DistSW#
```

Figure 3: DistSW-verify vlan brief

```
DistSW#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual up                    up
FastEthernet0/5        unassigned      YES manual up                    up
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
Vlan99                 192.168.46.194  YES manual up                    up
DistSW#
```

Figure 4: DistSW-verify trunk brief

4.2 Layer 3 Configuration

Router-on-a-stick was implemented on the CoreRouter's FastEthernet0/0 interface. Seven sub-interfaces were created, one per VLAN:

```
interface FastEthernet0/0.20
 description Gateway - Finance Dept
 encapsulation dot1Q 20
 ip address 192.168.46.65 255.255.255.224
 ip helper-address 192.168.46.178
 no shutdown
```

The **encapsulation dot1Q** command tags outbound frames with the correct VLAN identifier. The router performs full inter-VLAN routing over the single physical trunk.

```
CoreRouter>enable
CoreRouter#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  up                    up
FastEthernet0/0.10     192.168.46.1    YES manual up                    up
FastEthernet0/0.20     192.168.46.65   YES manual up                    up
FastEthernet0/0.30     192.168.46.97   YES manual up                    up
FastEthernet0/0.40     192.168.46.129  YES manual up                    up
FastEthernet0/0.50     192.168.46.161  YES manual up                    up
FastEthernet0/0.60     192.168.46.177  YES manual up                    up
FastEthernet0/0.99     192.168.46.193  YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
CoreRouter#
```

Figure 5: Router sub-interface brief

4.3 DHCP Scoped Multi-VLAN (Assigned Challenge)

The assigned challenge was to configure, verify, and demonstrate DHCP scoped multi-VLAN address assignment.

Architecture: A centralised DHCP server was placed in VLAN 60 (Server Farm) with a static IP of 192.168.46.178/28. Because DHCP uses broadcast and broadcasts do not cross VLAN boundaries the router was configured as a **DHCP Relay Agent** using the ip helper-address command on each user-VLAN sub-interface.

How DHCP Relay works:

1. A PC in VLAN 20 boots and broadcasts a DHCP DISCOVER
2. The broadcast stays within VLAN 20, it cannot reach VLAN 60
3. The router's Fa0/0.20 sub-interface intercepts the broadcast
4. Because ip helper-address 192.168.46.178 is configured, the router converts the broadcast to a unicast packet
5. The router forwards the unicast to the DHCP server
6. The server responds, and the router relays the response back to the PC

```
CoreRouter#show ip interface  Fa0/0.20
FastEthernet0/0.20 is up, line protocol is up (connected)
  Internet address is 192.168.46.65/27
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 192.168.46.178
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled

CoreRouter#
```

Figure 6: Router dhcp-relay

Table 4: DHCP Pool Configuration

Pool Name	Gateway	Start IP	Subnet Mask	Max Users
VOICE_POOL	192.168.46.1	192.168.46.10	255.255.255.192	50
FINANCE_POOL	192.168.46.65	192.168.46.70	255.255.255.224	25
CORP_POOL	192.168.46.97	192.168.46.102	255.255.255.224	25
COMM_POOL	192.168.46.129	192.168.46.134	255.255.255.224	25
MGT_POOL	192.168.46.161	192.168.46.166	255.255.255.240	10

Each pool's start IP is offset from the network address to reserve the first several addresses for the gateway and future static devices (printers, etc.).

The VOICE_POOL additionally includes a TFTP Server option set to 192.168.46.1 so that IP phones can locate the CallManager.

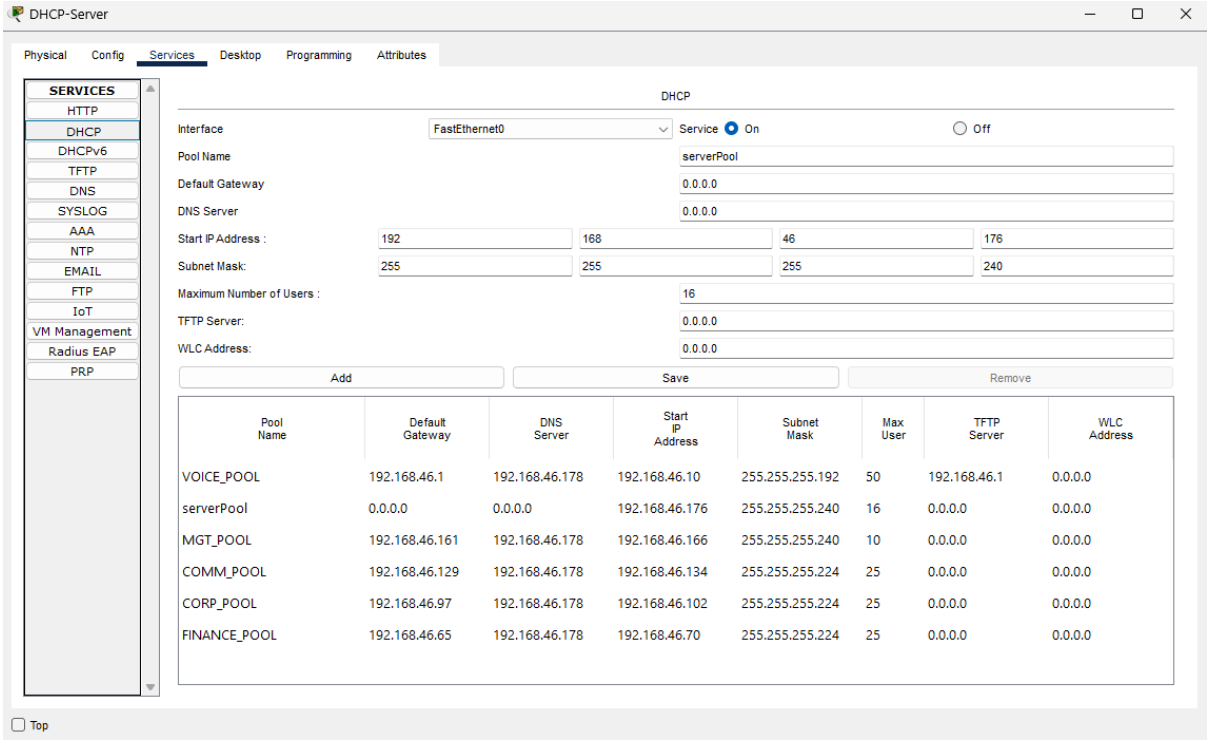


Figure 7: DHCP Pool configuration

4.4 VoIP Configuration (CR11)

Voice traffic was accommodated through:

1. **Dedicated Voice VLAN 10** across all departmental access switches
2. **Cisco 7960 IP phones** with built-in three-port switch PC and phone share one cable drop
3. **802.1Q tagging** phones tag their own frames into VLAN 10, PCs remain untagged in the department VLAN

CallManager Express was configured on CoreRouter:

```
telephony-service
 max-ephones 4
 max-dn 4
 ip source-address 192.168.46.1 port 2000
 auto assign 1 to 4
exit

ephone-dn 1
 number 1001
…
ephone-dn 4
 number 1004
```

The phones automatically registered with the CME, obtaining their configuration and extensions from the router. All four phones display their assigned extensions (1001 through 1004).

```
CoreRouter>enable
CoreRouter#show ephone

ephone-1 Mac:000A.4146.83B4 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:192.168.46.12 1025 7960   keepalive 43 max_line 2
 button 1: dn 1  number 1001 CH1   IDLE

ephone-2 Mac:0030.F251.9119 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:192.168.46.11 1025 7960   keepalive 43 max_line 2
 button 1: dn 2  number 1002 CH1   IDLE

ephone-3 Mac:0060.3E18.7B10 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:192.168.46.13 1025 7960   keepalive 43 max_line 2
 button 1: dn 3  number 1003 CH1   IDLE

ephone-4 Mac:0001.435B.B079 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:192.168.46.10 1025 7960   keepalive 43 max_line 2
 button 1: dn 4  number 1004 CH1   IDLE
CoreRouter#
```

Figure 8: CME registration showing all 4 phones REGISTERED



Figure 9: VoIP call test between Phone1 and Phone2

4.5 Security Considerations

VLAN separation: Each department operates in its own broadcast domain, isolating sensitive traffic.

Native VLAN 99 for trunks: Untagged management traffic uses a dedicated VLAN, preventing VLAN hopping attacks.

ACLs on server farm: Access to the server farm is restricted by extended ACLs, allowing only specific departments to reach the server on HTTP/HTTPS while blocking all other inter-VLAN access.

VLAN 99 for management: All switch management IPs are on a dedicated VLAN, inaccessible to user workstations.

5. Testing and Verification

5.1 Testing Methodology

Testing was performed systematically across seven categories:

1. **DHCP Assignment** - verify each VLAN's DHCP pool works
2. **DHCP Renewal** - verify continuous DHCP service
3. **Inter-VLAN Connectivity** - verify routing between departments
4. **DHCP Relay Verification** - confirm **ip helper-address** is present
5. **VoIP/CME Verification** - confirm phones registered
6. **Layer 2 Verification** - VLANs, trunks, management Ips
7. **Layer 3 Verification** - sub-interface status

5.2 Testing Matrix

Table 5: Testing Matrix Summary

Category	Tests	Passed	Failed
DHCP Assignment	5	5	0
DHCP Renewal	1	1	0
Inter-VLAN Connectivity	3	3	0
DHCP Relay Verification	2	2	0
VoIP/CME (CR11)	6	6	0
Layer 2 Verification	8	8	0
Layer 3 Verification	1	1	0
Total	**26**	**26**	**0**

The full testing matrix with per-test details is available in Appendix B and in **testing/testing-matrix.md** in the GitHub repository.

5.3 Key Test Results

DHCP Assignment: All eight PCs and all four IP phones received correct addresses from their respective pools. PC1 received 192.168.46.70 (Finance), PC3 received 192.168.46.102 (Corp), PC5 received 192.168.46.134 (Community), PC7 received 192.168.46.166 (Manager). Each phone received an address in the 192.168.46.10-.62 range.

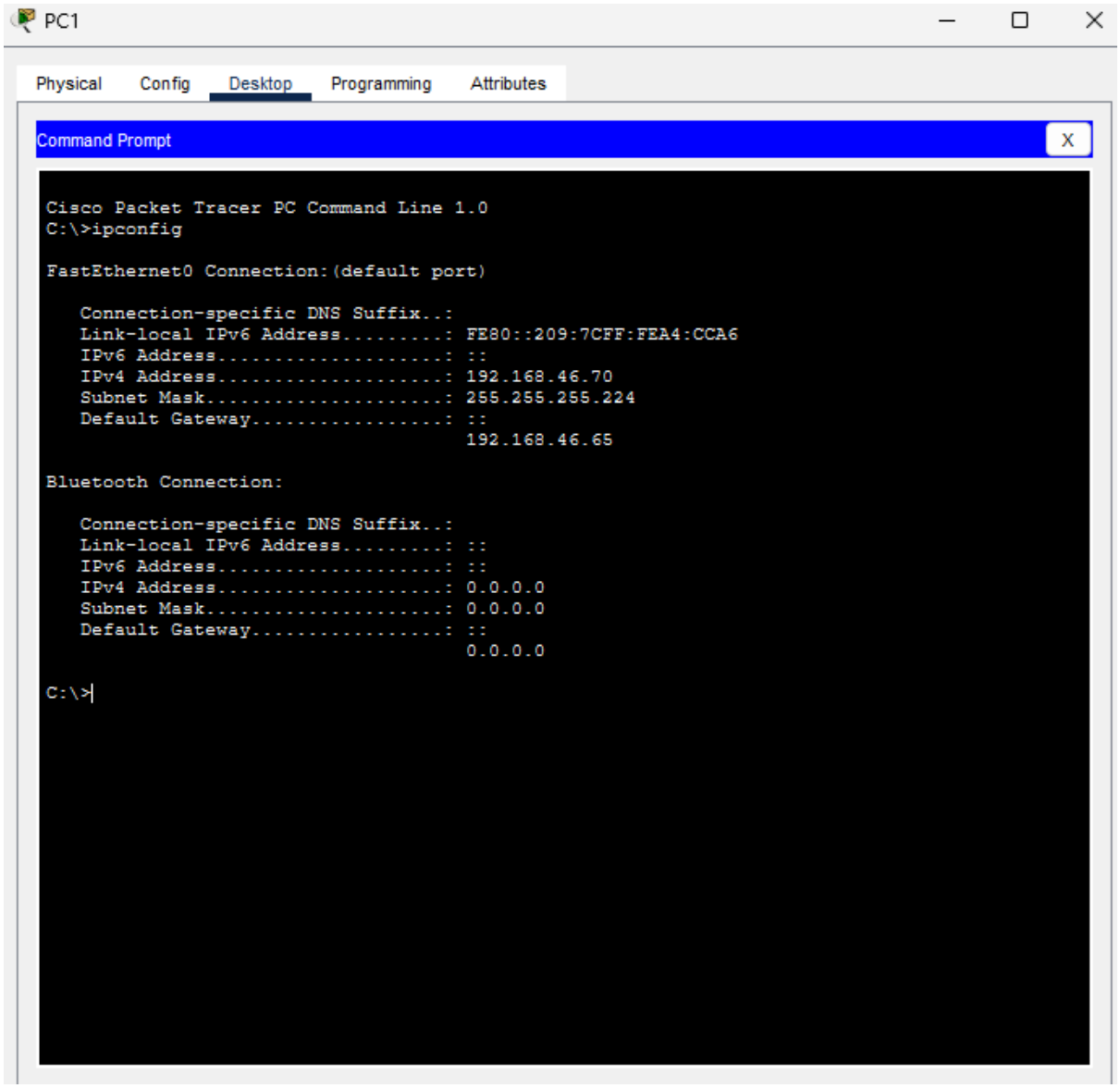


Figure 10: PC1 DHCP lease verification showing correct Finance pool assignment

Inter-VLAN Connectivity: PC1 successfully pinged its gateway (192.168.46.65), PC3 in a different VLAN (192.168.46.102), and the DHCP server (192.168.46.178). All pings returned 4/4 replies with 0% packet loss.

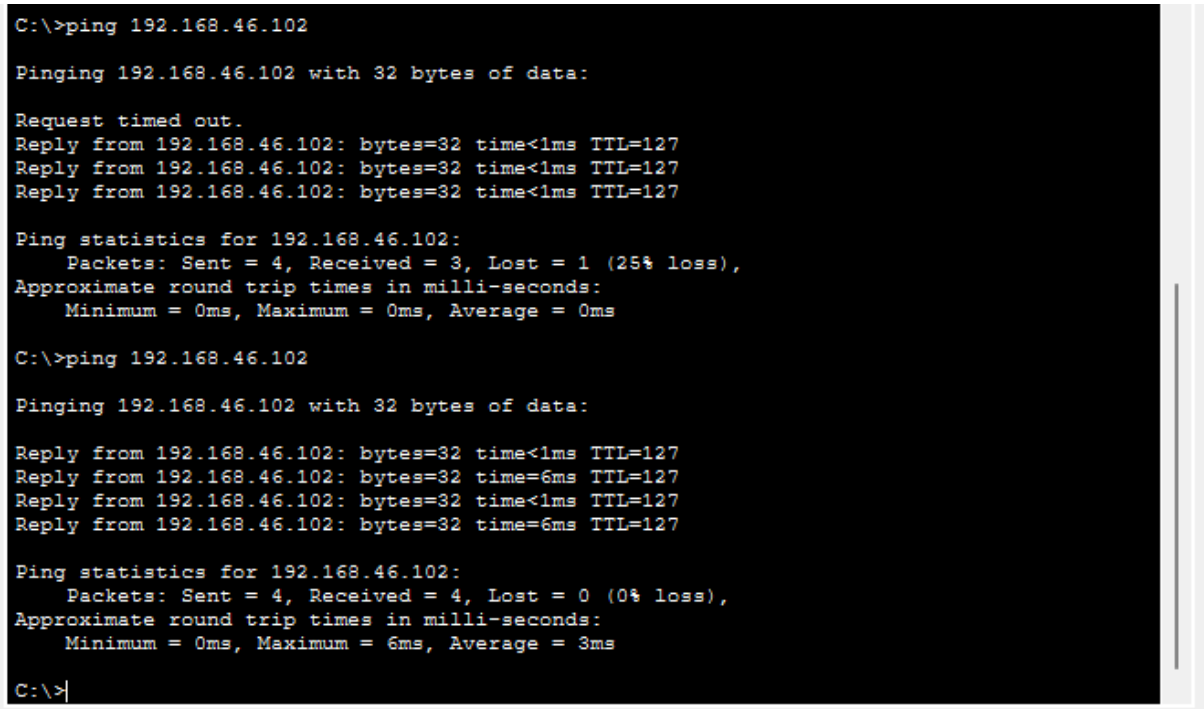


Figure 11: Inter-VLAN ping verification showing PC1 reaching PC2

VoIP Registration: The **show ephone** command on CoreRouter confirmed all four IP phones REGISTERED with extensions 1001 through 1004. A test call between Phone1 (1001) and Phone2 (1002) completed successfully.

DHCP Relay: The **show ip interface FastEthernet0/0.20** command confirmed the presence of **Helper address is 192.168.46.178** on the router's Finance sub-interface.

6. Troubleshooting

6.1 Summary of Issues

Table 6: Troubleshooting Summary

#	Issue	Resolution
1	Router uplink showed red triangle.	Issued no shutdown on the router interface.
2	Native VLAN mismatch warnings.	Completed native VLAN 99 config on all switch trunks.
3	VLAN99 SVI showed up/down.	Created VLAN 99 in the VLAN database on each switch.
4	**show running-config interface** returned invalid input.	Used **show ip interface** instead (Packet Tracer limitation)
5	**telephony-service** command unavailable on 2911.	Replaced 2911 with 2811 router model.
6	**show telephony-service** returned invalid.	Used **show ephone** and **show running-config** instead.
7	First ping attempt showed 25% packet loss.	Expected ARP cold-cache behaviour; no fix required.
8	Access switch trunks blocked DHCP traffic.	Expanded trunk allowed VLANs to include VLAN 60

6.2 Detailed Resolutions

Issue 1 - Router interface down: Unlike switch ports which come up automatically when cabled, router interfaces are administratively down by default. Running **no shutdown** on the interface resolved the issue.

Issue 2 - Native VLAN mismatch: When one end of a trunk is configured for native VLAN 99 while the other end still uses default VLAN 1, CDP generates mismatch warnings and spanning tree blocks the port. Completing native VLAN 99 on all trunks resolved the warnings.

Issue 3 - Vlan99 SVI down: The **interface vlan 99** command creates an SVI, but the SVI cannot come up unless VLAN 99 exists in the VLAN database. Running **vlan 99 → name MGMT_SWITCH** on each switch fixed this.

Issue 4 - Packet Tracer limitation: The command **show running-config interface GigabitEthernet0/0.20** is not implemented in Packet Tracer. Alternative: **show ip interface GigabitEthernet0/0.20**.

Issue 5 - Router model limitation: Packet Tracer only supports CallManager Express on the 2811 model. The router was swapped, and the router-on-a-stick config was reapplied using FastEthernet0/0.X sub-interfaces.

Issue 6 - Verification command unavailable: Packet Tracer does not implement **show telephony-service**. Alternative: **show ephone** and **show running-config | section telephony-service**.

Issue 7 - First-ping packet loss: When the ARP cache is empty, the first ICMP packet is dropped while ARP resolves the destination MAC address. This is expected and not a configuration error.

Issue 8 - DHCP blocked by restrictive trunk: Access switch trunks initially allowed only department VLANs and VLAN 99, blocking DHCP traffic to VLAN 60. Expanding allowed VLANs to include VLAN 60 resolved this.

The full troubleshooting log with diagnostic output is available in Appendix C and in **testing/troubleshooting-log.md**.

7. Reflection and Lessons Learned

7.1 Technical Lessons

This project demonstrated several important technical principles:

VLAN and trunk fundamentals: The importance of understanding that VLANs are local to each switch, the same VLAN must be created on every switch that carries it. Similarly, trunks must be configured symmetrically on both ends to avoid native VLAN mismatches and spanning tree blocks.

DHCP relay mechanics: The **ip helper-address** command is the essential mechanism for centralised DHCP in a segmented network. Without it, each VLAN would need its own DHCP server. Understanding how the router intercepts broadcasts and converts them to unicasts is a critical networking concept.

Router-on-a-stick design: A single physical interface can carry multiple VLANs using sub-interfaces. This is cost-effective for small networks but creates a single point of failure and bandwidth bottleneck.

Router model constraints: Packet Tracer restricts certain features to specific models. Choosing the correct hardware platform is as important as choosing the correct software configuration.

7.2 Design Lessons

The value of careful design: The VLSM addressing plan developed in Milestone 1 required no revisions during implementation. This validated the importance of thorough upfront planning.

Reserved capacity matters: The unassigned 192.168.46.208-.255 block provides genuine future-proofing for the municipality. Growth can occur without redesign.

Documentation as process: Maintaining a troubleshooting log throughout the project proved valuable several issues were resolved by referring back to previous diagnostic output.

7.3 Future Improvements

If this network were deployed in production:

- **High availability:** Add a second core router and redundant distribution links
- **Port security:** Enable sticky MAC addresses and 802.1X authentication on access ports
- **QoS policies:** Configure priority queuing for voice traffic on all trunk and router interfaces
- **DHCP snooping:** Prevent rogue DHCP servers from being attached to the network
- **Server redundancy:** Add a secondary DHCP and file server for fault tolerance

- **Logging and monitoring:** Configure Syslog and SNMP monitoring for proactive issue detection

8. Conclusion

The network design and implementation for the Kagisano-Molopo Local Municipality Offices successfully met every client requirement:

- **A working, tested network** with seven VLANs across five departments
- **The assigned DHCP scoped multi-VLAN challenge** fully implemented with five DHCP pools and functioning relay
- **The design constraint** for a future server addressed via a dedicated Server VLAN 60
- **The CR11 change request** for VoIP handsets fully accommodated through Voice VLAN 10 and CME with four registered extensions

All 26 verification tests passed. The Packet Tracer simulation reproduces a fully functional network that will serve as a reference for the actual deployment.

The design is scalable, secure, and well-documented, providing a solid foundation for the municipality's current operations and future growth.

9. References

- Cisco Systems, Inc. (2024). *Cisco IOS Configuration Fundamentals Command Reference*.
- Cisco Systems, Inc. (2024). *Cisco 2811 Integrated Services Router Data Sheet*.
- North-West University. (2026). *CMPG 325 Computer Networks Project Handbook*.
- North-West University. (2026). *Project Brief CMPG325-2026-103*.
- Cisco Networking Academy. (2024). *Introduction to Networks (CCNA v7.02)*.
- Odom, W. (2020). *Cisco CCNA 200-301 Official Cert Guide, Volume 1*. Cisco Press.
- DeepSeek AI
- Google Gemini AI

10. Appendices

Appendix A: Device Configurations

Full running configurations for all seven devices are available in the GitHub repository under **configuration/:**

- corerouter-config.txt
- distsw-config.txt
- finance-asw-config.txt
- corp-asw-config.txt
- community-asw-config.txt
- mgroffice-asw-config.txt
- server-asw-config.txt

Appendix B: Full Testing Matrix

See **testing/testing-matrix.md** in the GitHub repository for the complete 26-test matrix with per-test results.

Appendix C: Troubleshooting Log

See **testing/troubleshooting-log.md** in the GitHub repository for the complete troubleshooting log with diagnostic output.

Appendix D: GitHub Commit History

The GitHub repository is available at:
https://github.com/nqubekongcobo084-source/CMPG325-Semester-Project-2026.git

The commit history shows the complete development process from Milestone 1 through final submission.